熊子健 Zijian Xiong

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PROFILE

Ph.D Student in Huazhong University of Science and Technology, currently focusing on hardware software co-design of Compute-in-Memory architecture and memristor device.

EDUCATION

Huazhong University of Science and Technology, Wuhan, China — 2022—now
Ph.D student in Micro Electronics

University of Science and Technology of China, Hefei, China — 2018–2022
B.Sc. in Applied Physics (GPA: 3.62/4.30)

RESEARCH EXPERIENCE

1. **Lead Researcher, Nano-scale Energy and Computing Systems**
Huazhong University of Science and Technology

Project Description: This fundamental research project explored novel nanomaterials for two key applications: 1) generating sustainable energy from natural resources, and 2) developing next-generation, brain-inspired computing hardware for more efficient artificial intelligence.

Sustainable Energy Generation: Developed a system for generating clean energy from the natural mixing of fresh and salt water (salinity gradients) using silicon nano-channels.

Key Achievement: Achieved a power density of 705 W/m^2, demonstrating a highly efficient method for clean energy conversion.

Brain-Inspired Computing: Designed and fabricated a novel computing component (a memristor) using advanced 2D materials (h-BN) that mimics the function of a human brain synapse.

Key Achievement: Successfully demonstrated the device's ability to switch between short-term and long-term memory, a critical function for creating more efficient and powerful AI hardware for civilian applications.

Publication: Led the research and served as the first author for the resulting publication accepted by the IEEE Asia Conference on Power and Electrical Engineering (ACPEE), 2024.

PUBLICATION

Xiong, Z., Chen, K., Chen, Q., Miao, X.-S., & He, Y. (2024). "Advances in nano-channel energy harvesting and computing system". In *Proceedings of the IEEE Asia Conference on Power and Electrical Engineering (ACPEE)*.

Zhang, A., Li, D., Xu, L., **Xiong, Z.**, Zhang, J., Peng, H., & Luo, Q. (2022). "Simulation and optimization of a miniaturized ion source for a neutron tube". *Physical Review Accelerators and Beams*, 25(10), 103501.

Zhang, A., Xu, L., Sun, J., Peng, H., **Xiong, Z.**, Xiao, Y., & Luo, Q. (2022). "Thermal research of a single crystal tungsten target positron source for the STCF project in China". *Nuclear Instruments and Methods in Physics Research, Section A: Accelerators, Spectrometers, Detectors and Associated Equipment*, 1039, 167107.

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